

Question 27 (a)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the product rule	1

Sample answer:

$$f'(x) = e^{-2x} - 2xe^{-2x}$$

$$f''(x) = -2e^{-2x} - 2(x \times -2e^{-2x} + e^{-2x})$$

$$= -2e^{-2x} + 4xe^{-2x} - 2e^{-2x}$$

$$= (4x - 4)e^{-2x}$$

$$= 4(x - 1)e^{-2x}$$

Question 27 (b)

Criteria	Marks
• Provides correct solution	2
• Finds the x value of the stationary point	1

Sample answer:

Stationary points where $f'(x) = 0$

$$(1 - 2x)e^{-2x} = 0$$

$$x = \frac{1}{2}, \quad \text{since } e^{-2x} \neq 0, \quad f\left(\frac{1}{2}\right) = \frac{1}{2}e^{-1} = \frac{1}{2e}$$

$$f''\left(\frac{1}{2}\right) = 4 \times -\frac{1}{2} \times e^{-1} = -2e^{-1} < 0$$

$\therefore \left(\frac{1}{2}, \frac{1}{2e}\right)$ is a local maximum.

Question 27 (c)

Criteria	Marks
• Provides correct graph	3
• Sketches the curve showing most of the main features	2
• Provides a sketch showing the stationary point and 1 other feature	1

Sample answer:

Possible point of inflection where $f''(x) = 0$

$$4(x-1)e^{-2x} = 0$$

$$\therefore x = 1, \quad \text{since } e^{-2x} \neq 0$$

x	0	1	2
$f''(x)$	-4	0	$4e^{-4}$

Changes concavity at $x = 1$

$\therefore$ Inflection point at $x = 1$

$$\begin{aligned} f(1) &= 1e^{-2 \times 1} \\ &= e^{-2} \end{aligned}$$

$\left(1, \frac{1}{e^2}\right)$ — Inflection point

$f(0) = 0 \quad \therefore$ graph passes through $(0, 0)$

As $x \rightarrow \infty$, $f(x) \rightarrow 0$

